

StudyMate.AI

Vibe Coding: Building&DeployinganAIWebApplication on AWS

Project Concept Note & Comprehensive Development Report

LIVE AWS DEPLOYED APPLICATION URL:

<https://15.135.202.64.sslip.io>

Submitted for: Vibe Coding Masterclass Series
Organization: IBM & Bharat Cares

PART 1: PROJECT CONCEPT NOTE

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1. Project Title & Application Name

- Application Name: StudyMate.AI
- Project Title: Building & Deploying an AI-Powered Learning Assistant on AWS

2. Problem Statement & Objective

Students and researchers face cognitive overload when synthesizing dense study material, textbooks, and lecture notes. StudyMate-AI simplifies notes into ELI5 summaries, answers real-time queries via an AI tutor, and auto-generates MCQ practice quizzes.

3. Target User & Use Cases

- High School & University Students preparing for exams.
- Researchers & Educators analyzing documents (.pdf, .docx, .txt).
- Self-Learners seeking interactive subject guidance.

4. LLM Model & API Integration

- LLM Engine: Google Gemini 3.6 Flash
- API Integration: @google/genai Node.js SDK via secure Express.js backend endpoints.

5. Key Features

• Instant Notes Simplifier & ELI5 Generator

Converts raw notes/documents into concise structured bullet points.

• Multi-Format Document Ingestion

Parses PDFs, Word files (.docx), and plaintext (.txt).

• Interactive AI Tutor Chat

Real-time AI tutor offering step-by-step academic explanations.

• Automated MCQ Quiz & Score Analysis

Generates 5-question multiple choice quizzes with instant scoring.

6. Expected Outcomes & Live URL

Reduces study time by up to 60% and increases information retention.

Live AWS Application URL: <https://15.135.202.64.sslip.io>

PART 2: PROJECT DEVELOPMENT REPORT

Vibe Coding Methodology & AWS Deployment Summary

1. Tech Stack Overview

- Frontend: React 19, Vite, Tailwind CSS v4, Lucide Icons, Motion.
- Backend: Node.js 20, Express.js, Esbuild, Mammoth parser.
- AI Engine: Google Gemini 3.6 Flash (@google/genai SDK).
- Cloud: AWS EC2 (t2.micro / Ubuntu 24.04 LTS), Nginx Reverse Proxy, PM2, Let's Encrypt Certbot SSL.

2. Prompting Strategy & Frameworks

Persona Prompting

Configured AI with 'Expert Academic Exam Tutor' persona for clear, structured, encouraging responses.

JSON Schema Output

Enforced rigid JSON schema for Quiz Generation [{ id, question, options, correctIndex, explanation }].

3. Development Phases

Phase 1: Design & UI System

Implemented frosted-glass aesthetic with dark mode and mobile responsiveness.

Phase 2: Backend API Development

Built Express endpoints keeping Gemini API keys securely on server side.

Phase 3: Code Hardening & Git

Configured .gitignore to prevent secret leaks, pushing clean code to GitHub.

Phase 4: AWS Cloud Infrastructure

Provisioned EC2 instance, configured Security Groups (22, 80, 443), installed Node 20, PM2, and Nginx.

Phase 5: SSL Security & Optimization

Configured Let's Encrypt Certbot on 15.135.202.64.sslip.io and optimized mobile layout.

4. Challenges & Resolutions

Challenge: Secret Key Protection

GitHub Push Protection blocked commit due to API keys in env file. Resolved by clearing git history, updating .gitignore, and setting environment variables directly on EC2.

Challenge: Mobile Layout Overflow

Adjusted header navigation items and root container overflow-x-hidden for 390px mobile viewports.

5. Reflections & Conclusion

Demonstrated end-to-end Vibe Coding methodology—architecting, building, and deploying a functional AI web application on AWS Free Tier with zero cost and maximum security.